

Important: The portion highlighted in blue corresponds to the added proof.

1 Proof of Case 1

Case 1. Let t denote the window size. To derive τ^* , we start from the simple case $n_s < t$. Intuitively, a w -span segment incurs higher privacy loss when it is assigned to a window of size t than to a window of size $t + 1$. Therefore, we use the window size t to derive an upper bound on the privacy loss.

The conclusion is described below.

LEMMA 4.1. *Given a batch size K and a set of segments $\mathcal{S}$ obtained by partitioning the time series chronologically, let the window size $t = \lfloor |\mathcal{S}|/K \rfloor$ and suppose the number of w -span segments $n_s < t$. The maximum privacy loss is attained when the w -span segments are evenly allocated between two windows.*

PROOF. We analyze the privacy loss in two steps. **(1) Formulate the privacy loss.** Since the number of w -span segments n_s is smaller than the window size $t = \lfloor |\mathcal{S}|/K \rfloor$, the w -span segments are contained in at most two windows. Let x denote the number of w -span segments included in one window, the upper bound of privacy loss can be denoted as

$$\begin{aligned} \tau = & \frac{1}{\alpha - 1} \log\left(\frac{t-x}{t} \cdot \frac{t-(n_s-x)}{t} \exp((\alpha-1)D_\alpha^0)\right) \\ & + \left(\frac{x}{t} \cdot \frac{t-(n_s-x)}{t} + \frac{t-x}{t} \cdot \frac{n_s-x}{t}\right) \exp((\alpha-1)D_\alpha^1) \\ & + \frac{x}{t} \cdot \frac{n_s-x}{t} \exp((\alpha-1)D_\alpha^2). \end{aligned}$$

where D_α^m ($m = 0, 1, 2$) represents the Rényi divergence when m w -span segments are included. Since the sensitivity of the clipped gradient is $2C$ per sample, we add Gaussian noise with distribution $\mathcal{N}(0, 4C^2\sigma^2\mathbf{I})$. Let m be the number of selected samples from the w -span segments, B and B' be the batches from w -neighboring time series, and let f be the training model. The Rényi divergence can be bounded as

$$\begin{aligned} & D_\alpha(\mathcal{N}(f(B), 4C^2\sigma^2\mathbf{I}) \| \mathcal{N}(f(B'), 4C^2\sigma^2\mathbf{I})) \\ & = D_\alpha(\mathcal{N}(f(B) - f(B'), 4C^2\sigma^2\mathbf{I}) \| \mathcal{N}(0, 4C^2\sigma^2\mathbf{I})) \\ & \leq \sup_{\|\mathbf{v}\|_2 \leq 2mC} D_\alpha(\mathcal{N}(\mathbf{v}, 4C^2\sigma^2\mathbf{I}) \| \mathcal{N}(0, 4C^2\sigma^2\mathbf{I})) \\ & = \sup_{\|\mathbf{v}\|_2 \leq 2mC} \frac{\alpha \|\mathbf{v}\|_2}{2 \cdot 4C^2\sigma^2} = \frac{\alpha m^2}{2\sigma^2}. \end{aligned}$$

Therefore, the upper bound of privacy loss above can be derived as

$$\begin{aligned} \tau = & \frac{1}{\alpha - 1} \log\left(\frac{t-x}{t} \cdot \frac{t-(n_s-x)}{t}\right) \\ & + \left(\frac{x}{t} \cdot \frac{t-(n_s-x)}{t} + \frac{t-x}{t} \cdot \frac{n_s-x}{t}\right) \exp\left(\frac{\alpha(\alpha-1)}{2\sigma^2}\right) \\ & + \frac{x}{t} \cdot \frac{n_s-x}{t} \exp\left(\frac{4\alpha(\alpha-1)}{2\sigma^2}\right) \\ & = \frac{1}{\alpha - 1} \log\left(\text{Const} + \frac{1 - 2e^c + e^{4c}}{t^2} x(n_s - x)\right). \end{aligned} \tag{2}$$

where Const denotes the constant term and $c = \frac{\alpha(\alpha-1)}{2\sigma^2}$. **(2) Maximize the privacy loss for fixed α .** Let $a = e^c > 1$ and define $h(a) = a^4 - 2a + 1$. Then, we have $h(a) > h(1) = 0$ for all $a > 1$. Consequently, the maximized upper bound of privacy loss $\tau_{\max}$ is attained at $x = \lceil \frac{n_s}{2} \rceil$. $\square$

Compare the privacy costs across different window sizes. Intuitively, a w -span segment incurs higher privacy loss when it is assigned to a window of size t than when it is assigned to a window of size $t + 1$, since the sampling rate of a w -span segment is smaller in a window of size $t + 1$. Let $A = e^c - 1$, $B = e^{4c} - 2e^c + 1$, and $c = \frac{\alpha(\alpha-1)}{2\sigma^2}$. Let $F_{t,t}$ denote the logarithmic term of Eq. (2), $F_{t,t} = 1 + \frac{An_s}{t} + \frac{B}{t^2}x(n_s - x)$. The privacy loss $\tau_{t,t+1}$ of the two windows with size t and $t + 1$ is

$$\begin{aligned} \tau_{t,t+1} = & \frac{1}{\alpha - 1} \log\left(\frac{t-x}{t} \cdot \frac{t+1-(n_s-x)}{t+1}\right) \\ & + \left(\frac{x}{t} \cdot \frac{t+1-(n_s-x)}{t+1} + \frac{t-x}{t} \cdot \frac{n_s-x}{t+1}\right) \exp\left(\frac{\alpha(\alpha-1)}{2\sigma^2}\right) \\ & + \frac{x}{t} \cdot \frac{n_s-x}{t+1} \exp\left(\frac{4\alpha(\alpha-1)}{2\sigma^2}\right). \end{aligned}$$

Let $F_{t,t+1}$ denote the logarithmic term, $F_{t,t+1} = 1 + A(\frac{x}{t} + \frac{n_s - x}{t+1}) + \frac{B}{t(t+1)}x(n_s - x)$. Since $F_{t,t} - F_{t,t+1} = \frac{n_s - x}{t(t+1)}A + \frac{x(n_s - x)}{t^2(t+1)}B \geq 0$, it follows that $F_{t,t} \geq F_{t,t+1}$. Therefore, the maximum privacy loss is attained when the w -span segments are allocated to windows of size t .

2 Proof of Case 2

Case 2: $t \leq n_s < 2t$. We next consider the case $t \leq n_s < 2t$. In this regime, the w -span segments can occupy two or three windows. By Lemma 4.2, when the w -span segments are contained in two windows, the privacy loss is maximized when they are split evenly between the two windows. When they are contained in three windows, the privacy loss is maximized when one window is fully occupied by w -span segments and the remaining w -span segments are split evenly between the two windows adjacent to the fully occupied window. The corresponding maximum privacy loss is analyzed below.

LEMMA 4.2. Given a batch size K and a set of segments $\mathcal{S}$. Let $t = \lfloor |\mathcal{S}|/K \rfloor$ and suppose the number of w -span segments satisfies $t < n_s < 2t$. Define $c = \frac{\alpha(\alpha-1)}{2\sigma^2}$, where α is the order of Rényi divergence and σ is the noise multiplier of the Gaussian mechanism. Let $A = 1 - 3e^c + 3e^{4c} - e^{9c}$, $B = (2\rho - 1) + (2 - 4\rho)e^c + (2\rho - 1)e^{4c}$, $C = \left(\rho - \frac{1}{2}\right)^2 - \left(2\rho^2 + \frac{1}{2}\right)e^c + \left(\rho + \frac{1}{2}\right)^2 e^{4c}$, where $\rho \in \{0, \frac{1}{2t}, -\frac{1}{2t}\}$ is the discretization error induced by the ceiling operation. Define

$$b_0 = \begin{cases} b^*, & \text{if } B^2 - 4AC \geq 0 \text{ and } b^* = \frac{-B - \sqrt{B^2 - 4AC}}{2A} \in [0, \frac{1}{2}), \\ 0, & \text{otherwise.} \end{cases}$$

Then the privacy loss attains its maximum value:

- 1) If $n_s - t \leq 2\lceil b_0 t \rceil - 2$, the w -span segments are evenly split between two windows.
- 2) Otherwise, one window is first filled entirely with w -span segments and the remaining w -span segments are then equally split between the two boundary windows adjacent to the fully occupied window.

PROOF. We analyze the privacy loss case by case. **(1) Derive the privacy loss τ_2 when the w -span segments are distributed across two windows.** Let n_s be the number of w -span segments, K be the batch size, $t = \lfloor |\mathcal{S}|/K \rfloor$, and $D_\alpha^m = \frac{\alpha m^2}{2\sigma^2}$ ($m = 0, 1, 2, 3$) represent the Rényi divergence when m w -span segments are included in one batch. Let τ_2 denote the privacy loss when the w -span segments are contained in two windows,

$$\begin{aligned} \tau_2 = & \frac{1}{\alpha - 1} \log\left(\frac{(t-x)^2}{t^2} \exp((\alpha - 1)D_\alpha^0)\right) \\ & + 2 \cdot \frac{x}{t} \cdot \frac{t-x}{t} \exp((\alpha - 1)D_\alpha^1) + \frac{x^2}{t^2} \exp((\alpha - 1)D_\alpha^2), \end{aligned}$$

where $x = \lceil \frac{n_s}{2} \rceil$, and we have $\frac{x}{t} \in [0.5, 1)$. **(2) Derive the privacy loss τ_3 when the w -span segments are distributed across three windows.** Similarly, the privacy loss τ_3 when the w -span segments are contained in three windows can be denoted as

$$\begin{aligned} \tau_3 = & \frac{1}{\alpha - 1} \log\left(\frac{(t-y)^2}{t^2} \exp((\alpha - 1)D_\alpha^1)\right) \\ & + 2 \cdot \frac{y}{t} \cdot \frac{t-y}{t} \exp((\alpha - 1)D_\alpha^2) + \frac{y^2}{t^2} \exp((\alpha - 1)D_\alpha^3), \end{aligned}$$

where $y = \lceil \frac{n_s - t}{2} \rceil$, and hence $\frac{y}{t} \in [0, 0.5)$. **(3) Choose the maximum privacy loss between τ_2 and τ_3 .** Since $\log(\cdot)$ is a monotonically increasing function, it suffices to compare τ_2 and τ_3 by comparing their logarithmic terms. Let $b = \frac{y}{t} \in [0, 0.5)$ and denote $a = \frac{x}{t}$. For ease of analysis, we write $a = \frac{1}{2} + b + \rho$, where ρ denotes the discretization error induced by the ceiling operation and $\rho \in \{0, \frac{1}{2t}, -\frac{1}{2t}\}$.

The following explains $\rho \in \{0, \frac{1}{2t}, -\frac{1}{2t}\}$. Let

$$x = \left\lceil \frac{n_s}{2} \right\rceil, \quad y = \left\lceil \frac{n_s - t}{2} \right\rceil,$$

and define $a = \frac{x}{t}, b = \frac{y}{t}$, we can write

$$a = \frac{x}{t} = \frac{1}{2} + \frac{y}{t} + \rho = \frac{1}{2} + b + \rho,$$

where

$$\rho = \frac{x - y - \frac{t}{2}}{t}.$$

If $t = 2k$ is even, then $t/2 = k \in \mathbb{Z}$, and hence

$$x = \left\lceil \frac{n_s - t}{2} + k \right\rceil = \left\lceil \frac{n_s - t}{2} \right\rceil + k = y + k.$$

Therefore,

$$\rho = \frac{x - y - \frac{t}{2}}{t} = \frac{(y + k) - y - k}{t} = 0.$$

If $t = 2k + 1$ is odd, write

$$u = \frac{n_s - t}{2} = m + r, \quad m \in \mathbb{Z}, r \in [0, 1).$$

Then

$$y = \lceil u \rceil, \quad x = \left\lceil u + k + \frac{1}{2} \right\rceil = k + \left\lceil u + \frac{1}{2} \right\rceil.$$

Thus

$$x - y - \frac{t}{2} = \left\lceil u + \frac{1}{2} \right\rceil + k - \lceil u \rceil - \frac{1}{2} - k = \left\lceil u + \frac{1}{2} \right\rceil - \lceil u \rceil - \frac{1}{2}.$$

Now, according to the fractional part r of u ,

$$\left\lceil u + \frac{1}{2} \right\rceil - \lceil u \rceil - \frac{1}{2} = \left\lceil m + r + \frac{1}{2} \right\rceil - \lceil m + r \rceil - \frac{1}{2} = \begin{cases} \frac{1}{2}, & r = 0 \text{ or } \frac{1}{2} < r < 1, \\ -\frac{1}{2}, & 0 < r \leq \frac{1}{2}. \end{cases}$$

Hence,

$$\rho = \frac{x - y - \frac{t}{2}}{t} \in \left\{ \frac{1}{2t}, -\frac{1}{2t} \right\}.$$

Combining the even and odd cases yields

$$\rho \in \left\{ 0, \frac{1}{2t}, -\frac{1}{2t} \right\}.$$

We then get the logarithmic terms Lt_2 and Lt_3 from τ_2 and τ_3 , respectively.

$$\begin{aligned} Lt_2 &= (1 - a)^2 \exp((\alpha - 1)D_\alpha^0) + 2a(1 - a) \exp((\alpha - 1)D_\alpha^1) \\ &\quad + a^2 \exp((\alpha - 1)D_\alpha^2), \end{aligned}$$

and

$$\begin{aligned} Lt_3 &= (1 - b)^2 \exp((\alpha - 1)D_\alpha^1) + 2b(1 - b) \exp((\alpha - 1)D_\alpha^2) \\ &\quad + b^2 \exp((\alpha - 1)D_\alpha^3). \end{aligned}$$

By substituting $a = \frac{1}{2} + b + \rho$ into Lt_2 , we decompose the difference between Lt_2 and Lt_3 as $h(b) = Lt_2 - Lt_3$, yielding

$$\begin{aligned} h(b) &= \underbrace{(1 - 3e^c + 3e^{4c} - e^{9c})}_{A} b^2 \\ &\quad + \underbrace{((2\rho - 1) + (2 - 4\rho)e^c + (2\rho - 1)e^{4c})}_{B} b \\ &\quad + \underbrace{(\rho - \frac{1}{2})^2 - (2\rho^2 + \frac{1}{2})e^c + (\rho + \frac{1}{2})^2 e^{4c}}_{C}, \end{aligned}$$

where $c = \frac{\alpha(\alpha-1)}{2\sigma^2}$.

$$a = \frac{1}{2} + b + \rho$$

Substituting this into

$$Lt_2 = (1 - a)^2 + 2a(1 - a)e^c + a^2 e^{4c}, \quad Lt_3 = (1 - b)^2 e^c + 2b(1 - b)e^{4c} + b^2 e^{9c},$$

we derive the explicit expression of

$$h(b) = Lt_2 - Lt_3.$$

We first expand each term involving a . Since

$$1 - a = 1 - \left(\frac{1}{2} + b + \rho \right) = \frac{1}{2} - b - \rho,$$

we obtain

$$(1 - a)^2 = \left(\frac{1}{2} - b - \rho \right) \left(\frac{1}{2} - b - \rho \right) = \frac{1}{4} - \frac{1}{2}b - \frac{1}{2}\rho - \frac{1}{2}b + b^2 + b\rho - \frac{1}{2}\rho + b\rho + \rho^2 = b^2 - (1 - 2\rho)b + \left(\rho - \frac{1}{2} \right)^2.$$

Next,

$$a(1 - a) = \left(\frac{1}{2} + b + \rho \right) \left(\frac{1}{2} - b - \rho \right) = \frac{1}{4} - \frac{1}{2}b - \frac{1}{2}\rho + \frac{1}{2}b - b^2 - b\rho + \frac{1}{2}\rho - b\rho - \rho^2.$$

Therefore,

$$2a(1-a)e^c = \left(\frac{1}{2} - 2b^2 - 4\rho b - 2\rho^2\right)e^c.$$

For the last term,

$$a^2 = \left(\frac{1}{2} + b + \rho\right)^2 = b^2 + (1+2\rho)b + \left(\rho + \frac{1}{2}\right)^2,$$

and hence

$$a^2 e^{4c} = \left[b^2 + (1+2\rho)b + \left(\rho + \frac{1}{2}\right)^2\right] e^{4c}.$$

Combining the above expansions yields

$$\begin{aligned} Lt_2 &= (1-a)^2 + 2a(1-a)e^c + a^2 e^{4c} \\ &= \left[b^2 - (1-2\rho)b + \left(\rho - \frac{1}{2}\right)^2\right] \\ &\quad + \left(\frac{1}{2} - 2b^2 - 4\rho b - 2\rho^2\right)e^c \\ &\quad + \left[b^2 + (1+2\rho)b + \left(\rho + \frac{1}{2}\right)^2\right]e^{4c}. \end{aligned}$$

We now expand Lt_3 :

$$\begin{aligned} (1-b)^2 e^c &= (1-2b+b^2)e^c, \\ 2b(1-b)e^{4c} &= (2b-2b^2)e^{4c}. \end{aligned}$$

Thus,

$$Lt_3 = (1-2b+b^2)e^c + (2b-2b^2)e^{4c} + b^2 e^{9c}.$$

Hence,

$$\begin{aligned} h(b) &= Lt_2 - Lt_3 \\ &= \left[b^2 - (1-2\rho)b + \left(\rho - \frac{1}{2}\right)^2\right] \\ &\quad + \left(\frac{1}{2} - 2b^2 - 4\rho b - 2\rho^2\right)e^c \\ &\quad + \left[b^2 + (1+2\rho)b + \left(\rho + \frac{1}{2}\right)^2\right]e^{4c} \\ &\quad - (1-2b+b^2)e^c - (2b-2b^2)e^{4c} - b^2 e^{9c}. \end{aligned}$$

We now collect terms according to the powers of b .

Quadratic term. The coefficient of b^2 is

$$b^2 - 2b^2 e^c + b^2 e^{4c} - b^2 e^c + 2b^2 e^{4c} - b^2 e^{9c} = (1 - 3e^c + 3e^{4c} - e^{9c})b^2.$$

Linear term. The coefficient of b is

$$-(1-2\rho)b - 4\rho e^c b + (1+2\rho)e^{4c}b + 2e^c b - 2e^{4c}b,$$

which simplifies to

$$\left((2\rho-1) + (2-4\rho)e^c + (2\rho-1)e^{4c}\right)b.$$

Constant term. The constant term is

$$\left(\rho - \frac{1}{2}\right)^2 + \left(\frac{1}{2} - 2\rho^2\right)e^c + \left(\rho + \frac{1}{2}\right)^2 e^{4c} - e^c,$$

that is,

$$\left(\rho - \frac{1}{2}\right)^2 - \left(2\rho^2 + \frac{1}{2}\right)e^c + \left(\rho + \frac{1}{2}\right)^2 e^{4c}.$$

Therefore, we finally obtain

$$\begin{aligned} h(b) &= (1 - 3e^c + 3e^{4c} - e^{9c})b^2 \\ &\quad + \left((2\rho - 1) + (2 - 4\rho)e^c + (2\rho - 1)e^{4c} \right)b \\ &\quad + \left(\rho - \frac{1}{2} \right)^2 - \left(2\rho^2 + \frac{1}{2} \right)e^c + \left(\rho + \frac{1}{2} \right)^2 e^{4c}. \end{aligned}$$

Let $g(e^c) = 1 - 3e^c + 3e^{4c} - e^{9c}$ for $e^c > 1$, $g(e^c)$ is monotonically decreasing on $(1, \infty)$. Hence, we have $A < 0$. Similarly, we have $B < 0$. Recall that

$$B = (2\rho - 1) + (2 - 4\rho)e^c + (2\rho - 1)e^{4c}.$$

we factorize B as

$$\begin{aligned} B &= (2\rho - 1) - 2(2\rho - 1)e^c + (2\rho - 1)e^{4c} \\ &= (2\rho - 1)(1 - 2e^c + e^{4c}) < 0, \end{aligned}$$

for $\rho < \frac{1}{2}$.

Since $h(0.5) < 0$ holds for all $t > 6$, it follows that $Lt_2 < Lt_3$ when h has no root in $[0, 0.5)$. To facilitate subsequent analysis, we set $b_0 = 0$ when h has no positive solution.

Let $x = e^c$, $h(0.5) = \rho^2(1 - 2x + x^4) + 2\rho(x^4 - x) - \frac{1}{4}x^9 + \frac{1}{2}x^4 - \frac{1}{4}x$.

1) $\rho = 0$, $h(0.5) = -\frac{1}{4}(x^9 - 2x^4 + x)$. Let $g(x) = x^9 - 2x^4 + x$, $g'(x) = 9x^8 - 8x^3 + 1 > g'(1) = 0$, $g(x) > g(1) = 0$ for all $x > 1$. Therefore, $h(0.5) < 0$.

2) $\rho = \frac{1}{2t}$,

$$\begin{aligned} h\left(\frac{1}{2}\right) &= \frac{1}{4t^2} \left[(1 - 2x + x^4) + 4t(x^4 - x) + t^2(-x^9 + 2x^4 - x) \right] \\ &= \frac{x-1}{4t^2} \left[-t^2(x^8 + x^7 + x^6 + x^5 + x^4 - x^3 - x^2 - x) + 4t(x^3 + x^2 + x) + (x^3 + x^2 + x - 1) \right]. \end{aligned}$$

Let $\Phi(t) = [-t^2(x^8 + x^7 + x^6 + x^5 + x^4 - x^3 - x^2 - x) + 4t(x^3 + x^2 + x) + (x^3 + x^2 + x - 1)]$, we need $\Phi(t) < 0$.

Let $D = x^8 + x^7 + x^6 + x^5 + x^4 - x^3 - x^2 - x > 0$, $E = x^3 + x^2 + x > 0$, $F = x^3 + x^2 + x - 1 > 0$, we have $\Phi(t) = -Dt^2 + 4Et + F < 0$ when $t > \frac{2E + \sqrt{4E^2 + DF}}{D}$. Let $T(x) = \frac{2E + \sqrt{4E^2 + DF}}{D}$. To obtain the upper bound of $T(x)$, we need to find c that $\frac{2E + \sqrt{4E^2 + DF}}{D} \leq c$. Since $3D > 2E$ and $10D^2 > 4E^2 + DF$, we have $\frac{2E + \sqrt{4E^2 + DF}}{D} < 3 + \sqrt{10}$. Therefore, we need $t > 3 + \sqrt{10} > 6$.

3) $\rho = -\frac{1}{2t}$.

$$\begin{aligned} h\left(\frac{1}{2}\right) &= \frac{1}{4t^2} \left[(1 - 2x + x^4) - 4t(x^4 - x) + t^2(-x^9 + 2x^4 - x) \right] \\ &= -\frac{x-1}{4t^2} \left[t^2(x^8 + x^7 + x^6 + x^5 + x^4 - x^3 - x^2 - x) + 4t(x^3 + x^2 + x) + (-x^3 - x^2 - x + 1) \right] \\ &= -\frac{x-1}{4t^2} \left[t^2(x^8 + x^7 + x^6 + x^5 + x^4 - x^3 - x^2 - x) + (4t-1)(x^3 + x^2 + x) + 1 \right] < 0, \end{aligned}$$

since the term in $[\cdot]$ is positive.

Otherwise, $h = 0$ has exactly one solution in the interval $[0, 0.5)$, which is derived as $b_0 = \frac{-B - \sqrt{B^2 - 4AC}}{2A}$. Consequently, if $\lceil \frac{n_s - t}{2} \rceil < b_0 t$, namely, $n_s - t \leq 2\lceil b_0 t \rceil - 2$, we have $Lt_2 > Lt_3$.

If $\lceil \frac{n_s - t}{2} \rceil < b_0 t$, then, since the left-hand side is an integer, we have $\lceil \frac{n_s - t}{2} \rceil \leq \lceil b_0 t \rceil - 1$. Hence $\frac{n_s - t}{2} \leq \lceil \frac{n_s - t}{2} \rceil \leq \lceil b_0 t \rceil - 1$, which implies $n_s - t \leq 2\lceil b_0 t \rceil - 2$. $\square$

3 Proof of Sensitivity characterization of spaced sampling.

THEOREM 4.3 (SENSITIVITY CHARACTERIZATION OF SPACED SAMPLING.). *Given a batch size K and a set of segments $\mathcal{S}$ obtained by partitioning the time series chronologically, let $t = \lfloor |\mathcal{S}|/K \rfloor \geq 7$, and let n_s denote the number of w -span segments. Define $c = \frac{\alpha(\alpha-1)}{2\sigma^2}$, where α is the order of Rényi divergence and σ is the noise multiplier of the Gaussian mechanism under spaced sampling. Let $A = 1 - 3e^c + 3e^{4c} - e^{9c}$, $B = (2\rho - 1) + (2 - 4\rho)e^c + (2\rho - 1)e^{4c}$, $C = \left(\rho - \frac{1}{2}\right)^2 - \left(2\rho^2 + \frac{1}{2}\right)e^c + \left(\rho + \frac{1}{2}\right)^2 e^{4c}$, where $\rho \in \{0, \frac{1}{2t}, -\frac{1}{2t}\}$ is the discretization error induced by the ceiling operation. Define*

$$b_0 = \begin{cases} 0.5, & \text{if } n_s < t, \\ b^*, & \text{if } n_s \geq t, B^2 - 4AC \geq 0, \text{ and } b^* = \frac{-B - \sqrt{B^2 - 4AC}}{2A} \in [0, \frac{1}{2}), \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Then the privacy loss attains its maximum value when

$$\max \left(\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor, 0 \right) + 2 \text{ w-span segments are included in one batch.}$$

PROOF. In Case1, at most two neighboring windows can contain w-span segments. Hence, the maximum number of included w-span segments is 2. On the other hand, since $n_s < t$, we have $n_s - (2\lceil b_0 t \rceil - 1) < t$, which implies $\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor \leq 0$. Therefore, $\max \left(\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor, 0 \right) + 2 = 2$. The remaining two cases can be verified similarly.

Case 2: $t \leq n_s < 2t$. As shown above, let $b = \frac{1}{t} \lceil \frac{n_s - t}{2} \rceil$. If $b < b_0$, then $Lt_2 > Lt_3$, and the privacy loss is maximized when the w-span segments are evenly split between two windows. Otherwise, the privacy loss is maximized when one window is first filled with w-span segments and the remaining w-span segments are evenly split between the two adjacent windows.

Moreover, $\lceil \frac{n_s - t}{2} \rceil < b_0 t$ if and only if $n_s - t \leq 2\lceil b_0 t \rceil - 2$. Hence, if $n_s - t \leq 2\lceil b_0 t \rceil - 2$, the maximum number of included w-span segments is 2; otherwise, it is 3.

These two cases can be unified as $\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor + 2$. Indeed, since $t \leq n_s < 2t$, we have $0 \leq n_s - t < t$. If $n_s - t \leq 2\lceil b_0 t \rceil - 2$, then $n_s - (2\lceil b_0 t \rceil - 1) < t$, and thus $\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor = 0$, which gives 2. Otherwise, $n_s - t \geq 2\lceil b_0 t \rceil - 1$, and hence $n_s - (2\lceil b_0 t \rceil - 1) \geq t$. Since $n_s < 2t$, we also have $n_s - (2\lceil b_0 t \rceil - 1) < 2t$, so $\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor = 1$, which gives 3.

Case 3: $n_s \geq 2t$. Let $n = \lfloor \frac{n_s}{t} \rfloor$ and write $n_s = (n-1)t + (t+r)$, where $0 \leq r < t$. After filling $(n-1)$ windows completely, the remaining $t+r$ w-span segments fall into the regime of Case 2. Therefore, if $r \leq 2\lceil b_0 t \rceil - 2$, the remaining segments are evenly split between two windows, and the maximum number of included w-span segments is $(n-1) + 2 = n+1$. Otherwise, one window is first filled with w-span segments and the remaining segments are evenly split between the two adjacent windows, in which case the maximum number of included w-span segments is $n+2$.

These two subcases can again be unified. Since $n_s = nt + r$, we have $\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor = \left\lfloor \frac{nt + r - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor$. If $r \leq 2\lceil b_0 t \rceil - 2$, then $r - (2\lceil b_0 t \rceil - 1) < 0$, and thus $\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor = n-1$, which yields $\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor + 2 = n+1$. Otherwise, $r \geq 2\lceil b_0 t \rceil - 1$, and hence $0 \leq r - (2\lceil b_0 t \rceil - 1) < t$. Therefore, $\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor = n$, which gives $\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor + 2 = n+2$.

Combining the above three cases, the privacy loss attains its maximum value when $\max \left(\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor, 0 \right) + 2$ w-span segments are included in one batch. $\square$

4 Comparison with Poisson Sampling

THEOREM 4.4. Given batch size K and segment count $|S|$, let $\mathbb{E}_{\text{Poisson}} = n_s q = n_s \frac{K}{|S|}$ denote the expected number of sampled w-span segments under Poisson sampling. Then the number of included w-span segments under spaced sampling is upper bounded by $\frac{\mathbb{E}_{\text{Poisson}} |S|}{|S| - K} + 3$.

PROOF. The number of included w-span segments under spaced sampling is upper bounded by $\max \left(\left\lfloor \frac{n_s - (2\lceil b_0 t \rceil - 1)}{t} \right\rfloor, 0 \right) + 2 \leq \left\lfloor \frac{n_s + 1}{t} \right\rfloor + 2 \leq \frac{n_s + 1}{t} + 2$.

Since $t = \lfloor |S|/K \rfloor \geq |S|/K - 1$, we have $\frac{1}{t} \leq \frac{K}{|S| - K}$. Therefore, $\frac{n_s + 1}{t} + 2 \leq \frac{n_s K}{|S| - K} + 3 = \mathbb{E}_{\text{Poisson}} \frac{|S|}{|S| - K} + 3$, where $\mathbb{E}_{\text{Poisson}}$ denotes the expectation of included w-span segments under Poisson sampling, i.e., $\mathbb{E}_{\text{Poisson}} = n_s \cdot \frac{K}{|S|}$. $\square$

5 Comparison of Covariances

THEOREM 5.2. Let $\hat{g}_{\text{psp}}$ denote the resulting gradient estimator under Poisson spaced sampling, and let $\hat{g}_{\text{poi}}$ denote the gradient estimator under Poisson sampling with the same expected batch size. Then, we have

$$\text{Cov}(\hat{g}_{\text{poi}}) - \text{Cov}(\hat{g}_{\text{psp}}) \succeq 0.$$

PROOF. Let $\mathcal{S}$ be the set of time series segments, K be the batch size, and λ be the window scaling factor. Poisson spaced sampling partitions the time axis into $N = \lfloor K/\lambda \rfloor$ consecutive windows and lets $\mathcal{S}_k$ denote the set of segments falling into the k -th window with $k \in \{1, \dots, N\}$. For each window $k \in \{1, \dots, N\}$, let g_k denote the gradient of a segment sampled from $\mathcal{S}_k$ according to the within-window sampling rule. Define

$$\mu_k = \mathbb{E}[g_k], \quad \Sigma_k = \text{Cov}(g_k), \quad \mu = \frac{1}{N} \sum_{k=1}^N \mu_k.$$

Define the within-window and between-window covariances by

$$\Sigma_{WW} = \frac{1}{N} \sum_{k=1}^N \Sigma_k, \quad \Sigma_{BW} = \frac{1}{N} \sum_{k=1}^N (\mu_k - \mu)(\mu_k - \mu)^\top.$$

By the law of total covariance,

$$\text{Cov}(g) = \frac{1}{N} \sum_{k=1}^N \Sigma_k + \frac{1}{N} \sum_{k=1}^N (\mu_k - \mu)(\mu_k - \mu)^\top = \Sigma_{WW} + \Sigma_{BW}.$$

By construction, a segment gradient g drawn uniformly from the union of the N windows follows the mixture distribution that first chooses a window uniformly from $\{1, \dots, N\}$ and then draws a segment according to the within-window sampling rule. Hence,

$$\mu = \mathbb{E}[g] = \frac{1}{N} \sum_{k=1}^N \mu_k.$$

We now decompose $\text{Cov}(g)$ into within-window and between-window parts. Using the definition of covariance,

$$\text{Cov}(g) = \mathbb{E}[(g - \mu)(g - \mu)^\top].$$

For a sample drawn from window k , we write

$$g - \mu = (g_k - \mu_k) + (\mu_k - \mu).$$

Substituting this into the covariance expression and averaging over the N windows gives

$$\text{Cov}(g) = \frac{1}{N} \sum_{k=1}^N \mathbb{E}[(g_k - \mu_k) + (\mu_k - \mu)((g_k - \mu_k) + (\mu_k - \mu))^\top].$$

Expanding the product,

$$\begin{aligned} \text{Cov}(g) &= \frac{1}{N} \sum_{k=1}^N \mathbb{E}[(g_k - \mu_k)(g_k - \mu_k)^\top] \\ &\quad + \frac{1}{N} \sum_{k=1}^N \mathbb{E}[(g_k - \mu_k)(\mu_k - \mu)^\top] \\ &\quad + \frac{1}{N} \sum_{k=1}^N \mathbb{E}[(\mu_k - \mu)(g_k - \mu_k)^\top] \\ &\quad + \frac{1}{N} \sum_{k=1}^N \mathbb{E}[(\mu_k - \mu)(\mu_k - \mu)^\top]. \end{aligned}$$

Since $\mu_k - \mu$ is deterministic and $\mathbb{E}[g_k - \mu_k] = 0$, the two cross terms vanish:

$$\mathbb{E}[(g_k - \mu_k)(\mu_k - \mu)^\top] = \mathbb{E}[g_k - \mu_k](\mu_k - \mu)^\top = 0,$$

and similarly,

$$\mathbb{E}[(\mu_k - \mu)(g_k - \mu_k)^\top] = 0.$$

Therefore,

$$\begin{aligned} \text{Cov}(g) &= \frac{1}{N} \sum_{k=1}^N \text{Cov}(g_k) + \frac{1}{N} \sum_{k=1}^N (\mu_k - \mu)(\mu_k - \mu)^\top \\ &= \frac{1}{N} \sum_{k=1}^N \Sigma_k + \frac{1}{N} \sum_{k=1}^N (\mu_k - \mu)(\mu_k - \mu)^\top \\ &= \Sigma_{WW} + \Sigma_{BW}. \end{aligned}$$

(1) Analyze the variance of Poisson spaced sampling. To simplify the analysis, we ignore the effect of rounding and set λ as the probability of selecting a window. Under Poisson spaced sampling, one segment is drawn independently from each selected window. Let $I_k \sim \text{Bernoulli}(\lambda)$ be the inclusion indicator for window selection, let N denote the random batch size with $\mathbb{E}[\lambda N] = K$, and $\hat{g}_{\text{psp}} = \frac{1}{K} \sum I_k g_k$.

$$\begin{aligned} \text{Cov}(\hat{g}_{\text{psp}}) &= \text{Cov}\left(\frac{1}{K} \sum_{k=1}^N I_k g_k\right) = \frac{1}{K^2} \sum_{k=1}^N \text{Cov}(I_k g_k) \\ &= \frac{1}{K^2} \sum_{k=1}^N \mathbb{E}[(I_k g_k)(I_k g_k)^\top] - \mathbb{E}[I_k g_k] \mathbb{E}[I_k g_k]^\top \\ &= \frac{1}{K} \Sigma_{WW} + \frac{1-\lambda}{K} \Sigma_{BW} + \frac{1-\lambda}{K} \mu \mu^\top. \end{aligned}$$

$$\begin{aligned}
\text{Cov}(\hat{g}_{\text{psp}}) &= \text{Cov}\left(\frac{1}{K} \sum_{k=1}^N I_k g_k\right) \\
&= \frac{1}{K^2} \sum_{k=1}^N \text{Cov}(I_k g_k) \\
&= \frac{1}{K^2} \sum_{k=1}^N (\mathbb{E}[(I_k g_k)(I_k g_k)^\top] - \mathbb{E}[I_k g_k] \mathbb{E}[I_k g_k]^\top) \\
&= \frac{1}{K^2} \sum_{k=1}^N (\lambda \mathbb{E}[g_k g_k^\top] - \lambda^2 \mu_k \mu_k^\top) \\
&= \frac{1}{K^2} \sum_{k=1}^N (\lambda \text{Cov}(g_k) + \lambda \mathbb{E}(g_k) \mathbb{E}(g_k)^\top - \lambda^2 \mu_k \mu_k^\top) \\
&= \frac{1}{K^2} \sum_{k=1}^N (\lambda \text{Cov}(g_k) + \lambda(1-\lambda) \mu_k \mu_k^\top)
\end{aligned}$$

Since $\mu = \mathbb{E}[g] = \frac{1}{N} \sum_{k=1}^N \mu_k = \frac{1}{K} \sum_{k=1}^N \lambda \mu_k$ and $\Sigma_{BW} = \frac{1}{N} \sum_{k=1}^N (\mu_k - \mu)(\mu_k - \mu)^\top = \frac{1}{N} \sum_{k=1}^N \mu_k \mu_k^\top - \mu \mu^\top - \mu \mu_k^\top + \mu \mu^\top = \frac{1}{N} \sum_{k=1}^N \mu_k \mu_k^\top - \mu \mu^\top$, the equation above can be rewritten as

$$\begin{aligned}
&= \frac{1}{K} \Sigma_{WW} + \frac{1-\lambda}{K} (\Sigma_{BW} + \mu \mu^\top) \\
&= \frac{1}{K} \Sigma_{WW} + \frac{1-\lambda}{K} \Sigma_{BW} + \frac{1-\lambda}{K} \mu \mu^\top.
\end{aligned}$$

(2) Analyze the variance of Poisson sampling. Let $I_s \sim \text{Bernoulli}(q)$ be the inclusion indicator, let M denote the random batch size with $\mathbb{E}[M] = K$, and $\hat{g}_{\text{poi}} = \frac{1}{K} \sum_{s \in \mathcal{S}} I_s g_s$. where, conditional on M , the sampled gradients $\hat{g}_1, \dots, \hat{g}_M$ follow the distribution of g . Then,

$$\begin{aligned}
\text{Cov}(\hat{g}_{\text{poi}}) &= \frac{1}{K^2} \sum_{s \in \mathcal{S}} \text{Cov}(I_s g_s) = \frac{|\mathcal{S}|}{K^2} (q \text{Cov}(g) + q(1-q) \mu \mu^\top) \\
&= \frac{|\mathcal{S}|}{K^2} (q(\Sigma_{WW} + \Sigma_{BW}) + q(1-q) \mu \mu^\top) \\
&= \frac{1}{K} (\Sigma_{WW} + \Sigma_{BW}) + \frac{1-q}{K} \mu \mu^\top,
\end{aligned}$$

where we use $\text{Cov}(g) = \Sigma_{WW} + \Sigma_{BW}$.

$$\begin{aligned}
\text{Cov}(\hat{g}_{\text{poi}}) &= \text{Cov}\left(\frac{1}{K} \sum_{s \in \mathcal{S}} I_s g_s\right) = \frac{1}{K^2} \sum_{s \in \mathcal{S}} \text{Cov}(I_s g_s) \\
&= \frac{1}{K^2} \sum_{s \in \mathcal{S}} \mathbb{E}[(I_s g_s)(I_s g_s)^\top] - \mathbb{E}[I_s g_s] \mathbb{E}[I_s g_s]^\top
\end{aligned}$$

Since $I_s \in \{0, 1\}$, we have $I_s^2 = I_s$. Hence, $\mathbb{E}[(I_s g_s)(I_s g_s)^\top] = \mathbb{E}[I_s g_s g_s^\top] = \mathbb{E}[I_s] \mathbb{E}[g_s g_s^\top] = q \mathbb{E}[g g^\top] = q(\text{Cov}(g) + \mu \mu^\top)$. And $\mathbb{E}[I_s g_s] = \mathbb{E}[I_s] \mathbb{E}[g_s] = q \mu$. Therefore,

$$\begin{aligned}
&\frac{1}{K^2} \sum_{s \in \mathcal{S}} \mathbb{E}[(I_s g_s)(I_s g_s)^\top] - \mathbb{E}[I_s g_s] \mathbb{E}[I_s g_s]^\top \\
&= \frac{|\mathcal{S}|}{K^2} [q(\text{Cov}(g) + \mu \mu^\top) - q^2 \mu \mu^\top] \\
&= \frac{|\mathcal{S}|}{K^2} [q \text{Cov}(g) + q(1-q) \mu \mu^\top] \\
&= \frac{1}{K} (\Sigma_{WW} + \Sigma_{BW}) + \frac{1-q}{K} \mu \mu^\top.
\end{aligned}$$

Finally, we **compare the variance between Poisson spaced sampling and Poisson sampling**,

$$\text{Cov}(\hat{g}_{\text{poi}}) - \text{Cov}(\hat{g}_{\text{psp}}) = \frac{\lambda}{K} \Sigma_{BW} + \frac{\lambda - q}{K} \mu \mu^\top \succeq 0,$$

since Σ_{BW} is positive semidefinite and $\mu \mu^\top \succeq 0$. $\square$

6 Convergence Analysis

Convergence Analysis. We assume that the loss function f is L -smooth in time series training, and f is bounded below as $f^* = \inf_{\theta} f(\theta)$. Given the gradients g_i , the noisy clipped gradient of the t -th update is defined as $\tilde{g}_t = \frac{1}{K} \left(\sum_{i=1}^M \tilde{g}_i + z_t \right)$, where $z_t \sim \mathcal{N}(0, 4C^2\sigma^2\mathbf{I})$ and $\tilde{g}_i$ represent the clipped gradients. Let $\tilde{g}_t = \nabla f(\theta_t) + b_t + v_t$, where b_t denotes the bias induced by gradient clipping, and v_t is a zero-mean noise term capturing both the added Gaussian noise and the intrinsic randomness of the gradient estimator. We use a standard first-order measure, the average expected squared gradient norm, as a stationarity measure for nonconvex optimization. According to the Descent Lemma, the average expected squared gradient norm is derived with learning rate $\eta < \frac{1}{4L}$ (L represents the Lipschitz constant) as

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\theta_t)\|^2 &\leq \frac{4(f(\theta_0) - f^*)}{\eta T} + \frac{3}{T} \sum_{t=0}^{T-1} \mathbb{E} \|b_t\|^2 \\ &\quad + 2L\eta(\sigma_s + \sigma_N), \end{aligned}$$

where θ_t represents the parameters in the t -th update, $\sigma_N = \frac{4C^2\sigma^2}{K^2} \text{tr}(\mathbf{I})$, and $\sigma_s = \sup_t \text{tr}(\text{Cov}(\frac{1}{K} \sum_{i=1}^M g_i))$. Therefore, the convergence is related to the covariance of the sampling method. Since the covariance matrix is positive semidefinite, it follows that the σ_s under Poisson spaced sampling is smaller than that under Poisson sampling. Consequently, the convergence analysis suggests that Poisson spaced sampling may achieve faster convergence, whereas Poisson sampling may converge more slowly in this bound.

PROOF. Assume that f is L -smooth, namely

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2, \quad \forall x, y.$$

Let the parameter update be

$$\theta_{t+1} = \theta_t - \eta \hat{g}_t,$$

where the noisy clipped gradient estimator is defined as

$$\hat{g}_t = \frac{1}{K} \left(\sum_{i=1}^m \tilde{g}_i + z_t \right), \quad z_t \sim \mathcal{N}(0, 4C^2\sigma^2\mathbf{I}).$$

We decompose $\hat{g}_t$ as

$$\hat{g}_t = \nabla f(\theta_t) + b_t + v_t,$$

where b_t is the bias induced by gradient clipping, and v_t is the zero-mean fluctuation term capturing both the stochasticity of the gradient estimator and the injected Gaussian noise. Let $\mu_t = \mathbb{E}[\hat{g}_t \mid \theta_t]$. Then define $b_t = \mu_t - \nabla f(\theta_t)$, $v_t = \hat{g}_t - \mu_t$. We have

$$\mathbb{E}[v_t \mid \theta_t] = \mathbb{E}[\hat{g}_t - \mu_t \mid \theta_t] = \mathbb{E}[\hat{g}_t - \mathbb{E}[\hat{g}_t \mid \theta_t] \mid \theta_t] = 0.$$

By L -smoothness,

$$\begin{aligned} f(\theta_{t+1}) &\leq f(\theta_t) + \langle \nabla f(\theta_t), \theta_{t+1} - \theta_t \rangle + \frac{L}{2} \|\theta_{t+1} - \theta_t\|^2 \\ &= f(\theta_t) - \eta \langle \nabla f(\theta_t), \hat{g}_t \rangle + \frac{L\eta^2}{2} \|\hat{g}_t\|^2. \end{aligned}$$

Taking conditional expectation given θ_t ,

$$\mathbb{E}[f(\theta_{t+1}) \mid \theta_t] \leq f(\theta_t) - \eta \mathbb{E}[\langle \nabla f(\theta_t), \hat{g}_t \rangle \mid \theta_t] + \frac{L\eta^2}{2} \mathbb{E}[\|\hat{g}_t\|^2 \mid \theta_t]$$

As for the inner product term, we have

$$\begin{aligned} \mathbb{E}[\langle \nabla f(\theta_t), \hat{g}_t \rangle \mid \theta_t] &= \|\nabla f(\theta_t)\|^2 + \langle \nabla f(\theta_t), b_t \rangle + \mathbb{E}[\langle \nabla f(\theta_t), v_t \rangle \mid \theta_t] \\ &= \|\nabla f(\theta_t)\|^2 + \langle \nabla f(\theta_t), b_t \rangle, \end{aligned}$$

since $\mathbb{E}[v_t \mid \theta_t] = 0$. For the second moment term, we have

$$\begin{aligned} \mathbb{E}[\|\hat{g}_t\|^2 \mid \theta_t] &= \|\nabla f(\theta_t) + b_t\|^2 + 2 \langle \nabla f(\theta_t) + b_t, \mathbb{E}[v_t \mid \theta_t] \rangle + \mathbb{E}[\|v_t\|^2 \mid \theta_t] \\ &= \|\nabla f(\theta_t) + b_t\|^2 + \mathbb{E}[\|v_t\|^2 \mid \theta_t] \end{aligned}$$

$$\leq 2 \|\nabla f(\theta_t)\|^2 + 2\|b_t\|^2 + \mathbb{E}[\|v_t\|^2 \mid \theta_t].$$

Therefore,

$$\begin{aligned} \mathbb{E}[f(\theta_{t+1}) \mid \theta_t] &\leq f(\theta_t) - \eta \|\nabla f(\theta_t)\|^2 - \eta \langle \nabla f(\theta_t), b_t \rangle + L\eta^2 \|\nabla f(\theta_t)\|^2 + L\eta^2 \|b_t\|^2 + \frac{L\eta^2}{2} \mathbb{E}[\|v_t\|^2 \mid \theta_t] \\ &\leq f(\theta_t) - \eta \|\nabla f(\theta_t)\|^2 + \frac{\eta}{2} \|\nabla f(\theta_t)\|^2 + \frac{\eta}{2} \|b_t\|^2 + L\eta^2 \|\nabla f(\theta_t)\|^2 + L\eta^2 \|b_t\|^2 + \frac{L\eta^2}{2} \mathbb{E}[\|v_t\|^2 \mid \theta_t] \\ &= f(\theta_t) - \frac{\eta}{2} \|\nabla f(\theta_t)\|^2 + \frac{\eta}{2} \|b_t\|^2 + L\eta^2 \|\nabla f(\theta_t)\|^2 + L\eta^2 \|b_t\|^2 + \frac{L\eta^2}{2} \mathbb{E}[\|v_t\|^2 \mid \theta_t] \end{aligned}$$

Then, we have

$$\mathbb{E}[f(\theta_{t+1}) \mid \theta_t] \leq f(\theta_t) - \eta \left(\frac{1}{2} - L\eta \right) \|\nabla f(\theta_t)\|^2 + \left(\frac{\eta}{2} + L\eta^2 \right) \|b_t\|^2 + \frac{L\eta^2}{2} \mathbb{E}[\|v_t\|^2 \mid \theta_t].$$

Taking total expectation, we have

$$\mathbb{E}[f(\theta_{t+1})] \leq \mathbb{E}[f(\theta_t)] - \eta \left(\frac{1}{2} - L\eta \right) \mathbb{E}[\|\nabla f(\theta_t)\|^2] + \left(\frac{\eta}{2} + L\eta^2 \right) \mathbb{E}\|b_t\|^2 + \frac{L\eta^2}{2} \mathbb{E}\|v_t\|^2.$$

Let $\eta < \frac{1}{4L}$. Then, we have

$$\mathbb{E}[f(\theta_{t+1})] \leq \mathbb{E}[f(\theta_t)] - \frac{\eta}{4} \mathbb{E}[\|\nabla f(\theta_t)\|^2] + \frac{3\eta}{4} \mathbb{E}\|b_t\|^2 + \frac{L\eta^2}{2} \mathbb{E}\|v_t\|^2.$$

Summing over $t = 0, 1, \dots, T-1$, we have

$$\begin{aligned} \sum_{t=0}^{T-1} \mathbb{E}[f(\theta_{t+1})] &\leq \sum_{t=0}^{T-1} \mathbb{E}[f(\theta_t)] - \frac{\eta}{4} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(\theta_t)\|^2] + \frac{3\eta}{4} \sum_{t=0}^{T-1} \mathbb{E}\|b_t\|^2 + \frac{L\eta^2}{2} \sum_{t=0}^{T-1} \mathbb{E}\|v_t\|^2 \\ \sum_{t=0}^{T-1} \mathbb{E}[f(\theta_{t+1})] - \sum_{t=0}^{T-1} \mathbb{E}[f(\theta_t)] &\leq -\frac{\eta}{4} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(\theta_t)\|^2] + \frac{3\eta}{4} \sum_{t=0}^{T-1} \mathbb{E}\|b_t\|^2 + \frac{L\eta^2}{2} \sum_{t=0}^{T-1} \mathbb{E}\|v_t\|^2 \\ \mathbb{E}[f(\theta_T)] - \mathbb{E}[f(\theta_0)] &\leq -\frac{\eta}{4} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(\theta_t)\|^2] + \frac{3\eta}{4} \sum_{t=0}^{T-1} \mathbb{E}\|b_t\|^2 + \frac{L\eta^2}{2} \sum_{t=0}^{T-1} \mathbb{E}\|v_t\|^2 \\ \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(\theta_t)\|^2] &\leq \frac{4(f(\theta_0) - f^*)}{\eta T} + \frac{3}{T} \sum_{t=0}^{T-1} \mathbb{E}\|b_t\|^2 + 2L\eta T \Sigma^2, \end{aligned}$$

where $\Sigma^2 = \sup_t \mathbb{E}\|v_t\|^2$, and $f^* = \inf_{\theta} f(\theta)$. Since $\mathbb{E}[v_t] = 0$, $\mathbb{E}\|v_t\|^2 = \mathbb{E}\text{tr}(v_t v_t^\top) = \text{tr} \mathbb{E}(v_t v_t^\top) = \text{Cov}(v_t)$. Since v_t captures the stochasticity of the gradient estimator together with the injected Gaussian noise, we further bound

$$\Sigma^2 \leq \text{Cov}(v_t) = \sigma_s + \sigma_N,$$

where $\sigma_s = \sup_t \text{tr} \left(\text{Cov} \left(\frac{1}{K} \sum_{i=1}^M g_i \right) \right)$ denotes the variance contribution of the stochastic gradient estimator, and $\sigma_N = \frac{4C^2\sigma^2}{K^2} \text{tr}(I)$ is the variance of the injected Gaussian noise. Therefore,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(\theta_t)\|^2] \leq \frac{4(f(\theta_0) - f^*)}{\eta T} + \frac{3}{T} \sum_{t=0}^{T-1} \mathbb{E}\|b_t\|^2 + 2L\eta(\sigma_s + \sigma_N).$$

□